Saturable Vacuum Electrodynamics: a tree-level vacuum-birefringence prediction for a pump–probe X-ray polarimeter

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We register, before any pump-on data exists, a falsifiable prediction for the combined optical-pump / X-ray-probe polarimeter geometry of the BIREF@HIBEF proposal. The prediction follows from two constitutive statements: that the vacuum permittivity saturates as $\varepsilon_{\rm eff} = \varepsilon_0\, S(E)$ with the elliptic kernel $S = \sqrt{1-(E/E_{\rm c})^2}$ and one field scale $E_{\rm c} \simeq 1.13\times10^{17}\,{\rm V\,m^{-1}}$, and that the response keys on the field's sources — electric-field magnitude but magnetic *circulation* — a sector asymmetry that itself follows from the Maxwell source structure ($\nabla\cdot\mathbf{B} = 0$ admits no magnetic monopole to bias the response). Under a linearly polarized pump this yields a *tree-level*, E^2-leading birefringence $\delta n_{\rm bir} = n_\parallel - n_\perp \simeq -\frac{1}{2}(E/E_{\rm c})^2$ whose differential coefficient exceeds the one-loop QED (Euler–Heisenberg) prediction by the field-independent factor $\delta n_{\rm sat}/\delta n_{\rm QED} = 3.75\pi/\alpha^2 \simeq 2.2\times10^5$ (both coefficients on a single instantaneous footing). At the demonstrated ReLaX pump intensity ($10^{21}\,{\rm W\,cm^{-2}}$) the model predicts a polarization-flip probability $P_{\rm flip} \simeq 4\times10^{-3} - {-9}\times10^{-3}$, roughly seven orders of magnitude above the conservative (demonstrated) X-ray-polarimeter purity floor (2.4×10^{-10}). The model's magnetic response keys on circulation rather than static flux, so it predicts *zero* static-magnetic-field birefringence; existing PVLAS/BMV magnetic nulls are therefore consistent by construction, and the E-route configuration is the decisive one. We state the kill criterion explicitly: a pump-on polarimetric null (at 5σ) at or below the QED co-prediction, above the demonstrated purity floor and at $\gtrsim 10^{18}\,{\rm W\,cm^{-2}}$, falsifies the model's electric sector. Because the signal sits $\sim10^7$ above the demonstrated floor at already-demonstrated parameters, even commissioning-grade data adjudicates it. We report the coefficient honestly: the kernel *form* and the resulting coefficient *structure* are the model's content, whereas the absolute scale $E_{\rm c}$ is calibrated from the same measured constants (α, m_e) that fix the QED prediction.

I. INTRODUCTION

Quantum electrodynamics predicts that the vacuum, polarized by an intense electromagnetic field, becomes weakly birefringent [1, 2]. The effect is an α^2-suppressed one-loop response: for a probe crossing a field of strength E the two eigenmodes acquire an index difference $\delta n_{\rm QED} \sim \alpha^2(E/E_{\rm crit})^2$, with $E_{\rm crit} = m_e^2c^3/(e\hbar) \simeq 1.3\times10^{18}\,{\rm V\,m^{-1}}$ the Schwinger field. The predicted signal is minute and has never been measured directly. The BIREF@HIBEF proposal [3] targets it with a pump–probe geometry at the European XFEL: a petawatt-class optical laser (ReLaX) supplies the strong field and a polarized X-ray pulse probes the induced birefringence through a high-purity crossed polarimeter.

This Letter registers a competing, tree-level prediction for the *same* observable in the *same* apparatus. It rests on one constitutive hypothesis for the vacuum's dielectric response — that the permittivity *saturates* with field strength through a specific elliptic kernel — together with a companion statement keying the magnetic sector on circulation rather than static flux (Sec. II); we treat the pair here as a self-contained model of nonlinear electrodynamics. The model is not, however, interpretively neutral on the one point its sidereal falsifier turns on: it treats the vacuum as a real material medium possessing a rest frame, which we identify with the cosmic (CMB) rest frame (Sec. II). That premise underwrites the sidereal signature registered below — without a medium at rest in a definite frame the signature would vanish — and the model declares it openly rather than claiming agnosticism about it. We call it saturable vacuum electrodynamics (SVE). The model makes a numerically definite, pre-committed prediction that differs from QED by a fixed, field-independent factor, and it carries a distinctive signature — electric-field-active but static-magnetic-field-transparent — that separates it sharply from QED, which predicts a nonzero static-magnetic-field birefringence. Exact Born–Infeld likewise predicts zero static-magnetic-field *birefringence* — though for a distinct reason: among the zero-birefringence nonlinear electrodynamics — a family fully classified by Russo and Townsend [4] (Born–Infeld, Plebański, reverse and extreme Born–Infeld; the exceptional-case landscape sketched already in [5]) — exact B-I is the unique member *with a Maxwell weak-field limit* [6, 7], and it propagates both polarizations on a single *common* effective light cone (a common index shift with $\Delta n = 0$, not the full non-loading of the present model). A birefringence polarimeter measures Δn, so it cannot distinguish the two on the magnetic route; what then separates the model from exact B-I is its large tree-level electric-route coefficient, not the magnetic null.

The experimental landscape frames the test. Vacuum birefringence has been sought for decades on the magnetic route — most extensively by the PVLAS experi-

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ment [8], whose null bounds nonlinear-electrodynamics parameters [9, 10] but, as we show, cannot constrain the present electric-route model. The QED signal for the XFEL pump–probe geometry has been developed in detail [11, 12], and the required high-purity X-ray polarimetry has been demonstrated with the pump off [12, 13]. The present model is distinct from these alternatives on two counts: it predicts a *tree-level*, $O(1)$ electric-route coefficient (versus QED's one-loop signal and versus exact Born–Infeld's *zero*), and it predicts zero static-magnetic-field birefringence (versus QED; exact B-I predicts a zero magnetic Δn too — there via a common index shift, here via full non-loading — so a Δn polarimeter cannot separate the model from exact B-I on the magnetic route, and the electric-route coefficient, not the magnetic null, is what does). It is also distinct from a pseudoscalar (axion-like) signal [10]: the model produces pure retardance with no parity-odd polarization rotation on the electric route, whereas an axion coupling would rotate the plane of polarization; the two are separable by the analyzer geometry.

Our aim is a clean falsifier, not a defense. We therefore (i) state the model as a two-parameter constitutive law with its one calibrated scale ledgered plainly; (ii) compute the prediction and the QED co-prediction through an identical readout chain; (iii) show that no published experiment has yet combined a strong optical pump, a polarized X-ray probe, and a polarization-flip analysis, so this is a forward prediction rather than a retrodiction; and (iv) commit, before data, to the measurement outcome that would kill the model.

II. THE CONSTITUTIVE MODEL

A. The saturable permittivity

We postulate that the vacuum responds to an applied electric field as a dielectric whose permittivity *saturates* rather than remaining constant. Concretely, with ε_0 the low-field vacuum permittivity, we take the effective permittivity along the field to be

$$\varepsilon_{\rm eff}(E) = \varepsilon_0\, S(E), \qquad S(E) = \sqrt{1-\left(\frac{E}{E_{\rm c}}\right)^2}, \tag{1}$$

a single elliptic (circular-arc) kernel controlled by one field scale $E_{\rm c}$. The functional form is the defining hypothesis: S is unity at low field, softens as E grows, and reaches zero at $E = E_{\rm c}$, where the medium ceases to support propagation. Equation (1) is the only nonlinearity in the model; everything below is its consequence. We take S to be a function of the lab-focus-frame field magnitude $|E|$; the single-invariant $|E|^2$ collapse (in contrast to QED's two Lorentz invariants) is a modeling choice, and its frame dependence is addressed in Sec. II below.

For a linearly polarized pump field $\mathbf{E}_0$ the response becomes uniaxial, with optic axis along $\hat{\mathbf{E}}_0$. Writing $A \equiv E/E_{\rm c}$, the probe-response tensor is the exact differential of the scalar kernel, $\varepsilon_{ij} = \varepsilon\,\delta_{ij} + 2\varepsilon' E_{0i}E_{0j}$ with $\varepsilon' = {\rm d}\varepsilon/{\rm d}(E^2)$. The two refractive eigen-indices for a probe are

$$n_\perp = (1-A^2)^{1/4}, \qquad n_\parallel = \sqrt{\frac{1-2A^2}{\sqrt{1-A^2}}}, \tag{2}$$

and the birefringence a polarimeter measures is their difference,

$$\boxed{\delta n_{\rm bir} = n_\parallel - n_\perp \;\simeq\; -\tfrac{1}{2}A^2 = -\tfrac{1}{2}\left(\frac{E}{E_{\rm c}}\right)^2} \tag{3}$$

to leading order in A. This is negative, E^2-leading, and $O(1)$ against the saturation scale $E_{\rm c}$ — it is not loop-suppressed. It vanishes at low field ($A \to 0$), recovering an unbirefringent tree-level vacuum exactly as QED does at tree level; the distinction from QED is one of *coefficient*, at the same leading power (Sec. III). Every coefficient in this paper follows from Eq. (1) by differentiation — the tensor structure, both eigen-indices, the differential and common-mode shifts, and the ratio of Sec. III; no additional parameter or coupling enters anywhere (Appendix A).

B. The one calibrated scale

The kernel (1) has one dimensionful parameter, the saturation field $E_{\rm c}$. We fix it as

$$E_{\rm c} = \sqrt{\alpha}\, E_{\rm crit}, \qquad E_{\rm crit} = \frac{m_e^2 c^3}{e\hbar}, \tag{4}$$

which evaluates to $E_{\rm c} \simeq 1.13 \times 10^{17}\,{\rm V\,m^{-1}}$ using the fine structure constant α and the electron mass m_e. Equation (4) places the saturation field a factor $\sqrt{\alpha}$ below the QED Schwinger field $E_{\rm crit} \simeq 1.3 \times 10^{18}\,{\rm V\,m^{-1}}$, and it is the origin of the field-independent ratio to QED derived below, through the exact identity

$$\left(\frac{E_{\rm crit}}{E_{\rm c}}\right)^2 = \frac{1}{\alpha}. \tag{5}$$

Honesty ledger. We separate what the model supplies from what it imports, and we hold it to the same standard as QED. *(i) The kernel form is the model's content.* That the vacuum saturates at all — possessing a tree-level, $O(1)$, birefringence-bearing structure — is the substantive, potentially wrong hypothesis. The QED vacuum has no such tree-level structure; its birefringence is an α^2-loop effect. The existence and functional form of Eq. (1) is therefore the distinguishing physics. *(ii) The coefficient structure is the model's content.* The factor $\frac{1}{2}$ in Eq. (3) and $\sqrt{\alpha}$ in Eq. (4), and hence the α^{-2} scaling of the ratio to QED, follow from the form once $E_{\rm c}$ is fixed. *(iii) The absolute scale is calibrated, not derived.* The number $E_c \simeq 1.13 \times 10^{17}\,\mathrm{V\,m^{-1}}$ is set through Eq. (4) from the *same* measured constants (α, m_e) that every treatment of this problem imports; the model does not predict α. The resulting magnitude of the ratio to QED, $3.75\pi/\alpha^2$, is thus rooted in α in exactly the way QED's own one-loop coefficient $\sim \alpha$ is: neither framework derives α. We therefore do not present the numerical magnitude as the discriminating claim. The discriminating claim is the *existence of an $O(1)$ tree-level differential coefficient* where QED has only a loop-suppressed one — and this is what the experiment tests. *(iv) The single-footing convention, stated explicitly.* The QED Euler–Heisenberg coefficient carries a temporal-averaging convention that must be matched to the model's. The literature headline $\alpha/(15\pi)$ [Eq. (8)] is the pump's optical carrier *cycle-averaged* value ($\langle\cos^2\rangle = \frac{1}{2}$); its instantaneous (peak-field) value is twice that, $2\alpha/(15\pi)$. The model kernel $\delta n_{\rm bir} = -\frac{1}{2}(E/E_c)^2$ is, by contrast, an instantaneous algebraic response evaluated at the peak field (Appendix A). We therefore state the discriminating ratio on a *single instantaneous footing* — instantaneous model kernel against *instantaneous* one-loop coefficient $2\alpha/(15\pi)$ — giving $\delta n_{\rm sat}/\delta n_{\rm QED} = 3.75\pi/\alpha^2$ (Sec. III). Pairing instead the instantaneous kernel against the cycle-averaged $\alpha/(15\pi)$ is a mixed footing and doubles the quoted ratio to $7.5\pi/\alpha^2$; the two differ by exactly the $\langle\cos^2\rangle = \frac{1}{2}$ carrier average, and neither the order of magnitude nor any falsifier verdict of Sec. V depends on the choice. (This same $\frac{1}{2}$ carrier average is one of the two factors in the static-to-propagating QED normalization step of Sec. III.) *(v) The sector boundary was acquired, not assumed.* We record where the model failed and what that costs. *What was implicitly claimed:* the kernel of Eq. (1), written as a constitutive law with no regime boundary, reads as a universal permittivity for any electric field, static fields included. *What the data ruled:* that reading is dead. Run as a continuum static law down to atomic scales, the same kernel overshoots the muonic-hydrogen Lamb interval [14] non-perturbatively — the induced level shift exceeds the entire measured splitting, not merely its error bar (and inside a high-Z ion the kernel has no real solution at all) — with no physically-motivated short-distance cutoff to rescue it. Static fields are therefore placed *outside* the model's scope: the model asserts the constitutive law of Eq. (1) for the radiative (pump–probe) sector only, and makes no claim about the static-field permittivity. The muonic-hydrogen computation is accordingly a *domain-restriction demonstration*, not a static-field prediction — it exhibits the boundary the model must respect, and the model respects it by not extending there (Sec. II, sector scope). *What survives:* the falsifier of this Letter, unchanged. The registered prediction — $P_{\rm flip}$ and the kill criterion of Sec. V — is a *radiative*-sector quantity (a propagating, weak-field, time-varying pump crossed by a probe), disjoint from the atomic static-DC sector the model does not claim; the static-**B** transparency of Eq. (7) is disjoint again. Neither is touched by the domain restriction. *What is open:* the *mechanism* that scopes the law to the radiative sector. That the law is radiative-sector is here an *open postulate* — we state *that* the constitutive law holds for the radiative sector, not yet *why* it does not extend to the static one. The dynamical scoping routes proposed to derive that boundary — keys that would distinguish a propagating from a held field by its dynamical content, together with the weaker escape that a held static field might load the permittivity only slightly, cut off at a short-distance scale — have each been tested and excluded; static fields do not sit inside the model as a small effect, they sit outside it. We flag this openly rather than present the constitutive law as though it carried a settled sector boundary from the outset.

The elliptic form is not a free choice. The kernel $S = \sqrt{1-(E/E_c)^2}$ is the unique response of a lossless oscillator whose amplitude is confined to a hard bound under energy conservation. If the two reactive stores exchange a conserved energy while the drive amplitude E and the residual stiffness S are complementary components of a fixed-length state ($E^2 + E_c^2 S^2 = E_c^2$, the Pythagorean complement — as for a saturating inductor at its flux ceiling, or the frequency pull of a pendulum swinging to a hard stop), then $S = \sqrt{1-(E/E_c)^2}$ follows and no other exponent does: a general power law $S = (1-(E/E_c)^p)^{1/p}$ satisfies the same endpoints and small-field limit for *every* p, and only the quadratic (energy) norm $p = 2$ is selected by energy conservation. The exponent is therefore a consequence of the confinement hypothesis, not a fitted parameter. And it is testable in its own right: different kernel exponents predict different ratios of the anisotropic (par−perp) to the common-mode index shift, so measuring both channels fixes p — a second-order discriminator beyond the leading coefficient of Sec. III.

Family context. A saturating-field electrodynamics is not a new idea: Born and Infeld introduced one ninety years ago [6], engineering a square-root Lagrangian $\mathcal{L} \propto \sqrt{1+F/b^2}$ — with $F \equiv B^2 - E^2$ (the Lorentz scalar, $= \frac{1}{2}F_{\mu\nu}F^{\mu\nu}$ in Heaviside–Lorentz units) and a limiting field b — so that the field of a point charge — and the electron self-energy — stays finite. The present kernel $S = \sqrt{1-(E/E_c)^2}$ belongs to the same saturating-field family but is a distinct branch of it, and the cleanest way to state the difference is at the level of the response, not the Lagrangian: the two theories put the root on *opposite sides* of the electric response. In Born–Infeld the response *stiffens* — the displacement carries the root in its denominator, $D = E/\sqrt{1-(E/b)^2}$ (exponent $-\frac{1}{2}$), so D diverges as $E \to b$; here the permittivity *softens* — $\varepsilon = \varepsilon_0\sqrt{1-(E/E_c)^2}$ carries the root in the numerator (exponent $+\frac{1}{2}$), so $\varepsilon \to 0$ at the ceiling E_c. This response-exponent contrast is convention-independent, whereas comparing the sign of the invariant under the root is not: with $F \equiv B^2 - E^2$ a pure electric field gives $F = -E^2$, so the covariant and electric forms coincide exactly ($\sqrt{1+F/b^2} = \sqrt{1-E^2/b^2}$), and an interior-sign comparison is a display convention rather than a physical distinction. And (Sec. IV) the magnetic sector is not symmetric with the electric one. We situate the model here only to place its lineage; any comparison of self-energy properties across branches is outside the scope of this Letter, whose registered claim is the pump–probe birefringence coefficient.

Sector scope, stated explicitly. We assert the constitutive law of Eq. (1) for the *radiative* (pump–probe) sector that carries the registered prediction: a quiescent (unbiased, unexcited-vacuum), weak-field ($A^2 \simeq 6 \times 10^{-7}$), time-varying dielectric response read at optical/X-ray frequencies. A held *static* field lies *outside* this scope: the model makes no claim about the static-field permittivity. That boundary was not assumed away but forced by a pre-registered test. Before any external submission, we pre-registered a fork with a committed-in-advance outcome and computed the elliptic kernel's contribution, *were* it (wrongly) extended as a continuum static constitutive law, to the muonic-hydrogen Lamb shift. An atomic Coulomb field is not weak against E_c (at the muonic-hydrogen Bohr scale $A^2 \simeq 2.5 \times 10^{-2}$, and inside a high-$Z$ ion the field exceeds E_c, where the elliptic kernel has no real solution at all), and the muonic-hydrogen $2P_{1/2}$–$2S_{1/2}$ interval is measured to 202.3706(23) meV [14]. The continuum static extension *fails this test non-perturbatively*: the induced level shift exceeds not merely the sub-μeV measurement window but the *entire* measured Lamb interval by orders of magnitude, and no short-distance cutoff placed at a physically motivated scale — in particular at the node scale $\ell_{\rm node} \equiv \hbar/(m_e c) \simeq 3.86 \times 10^{-13}$ m, the electron's reduced Compton wavelength (the model's short-distance "node" spacing, below which the continuum form of the constitutive law is not guaranteed to be the correct accounting) — rescues it. This is a *domain-restriction demonstration*: it exhibits why the static-field extension of Eq. (1) cannot be a universal continuum law, and the model accordingly does not assert one. The registered radiative-sector prediction of this Letter is a separate sector, untouched by the restriction (honesty ledger, Sec. II).

A moot ordinary-matter bound. Because static fields are outside the model, no laboratory static-optics datum tests it either way; for orientation only, we record that even the (excluded) continuum static extension would be unobservable in ordinary matter. Volume-averaged over the copper unit cell with the short-distance cutoff taken at the copper-scale radius $r_c(\mathrm{Cu}) = \sqrt{Z} \times 160$ fm $\simeq$ 861 fm ($Z = 29$; the cutoff at which the radial integral is actually terminated — *not* the node scale $\ell_{\rm node} \simeq 386$ fm, which an earlier draft mis-named here), the X-ray refractive decrement the extended law would add to an ~ 8 keV probe in copper is $\delta_{\rm AVE} \simeq 4.2 \times 10^{-8}$, roughly three orders of magnitude below the measured copper decrement ($\delta \simeq 2.4 \times 10^{-5}$) and far beneath its known accuracy. This bound is moot for the registered claim; we keep it only to show the domain restriction hides no ordinary-matter tension.

The static extension also fails an X-ray transport test, at the opposite end of the size parameter from this copper bound. That a continuum static extension is excluded by high-Z X-ray transport, yet would add only an unobservable decrement to copper, is not a tension: the two are opposite limits of one dimensionless size parameter $x \equiv k\, r_c(Z)$, with k the probe wavenumber and $r_c(Z) = \sqrt{Z} \times 160$ fm the charge-scale radius of the anomalous halo the extension would place around each nucleus. That halo becomes optically resolved above the threshold $E_{\rm res} \equiv \hbar c/r_c(Z) \propto 1/\sqrt{Z}$, which *falls* with atomic number (higher Z resolves at lower energy). *Scattering channel — the exclusion.* For $Z \gtrsim 26$ and $E \gtrsim E_{\rm res}$ the halo is resolved ($x \gtrsim 1$) and its anomalous Born scattering would exceed the measured photon cross-section, so an ordinary high-Z transmission measurement in the 130–500 keV band already excludes the continuum static extension:

	Z	$r_c(Z)$ (fm)	$E_{\rm res}$ (keV)
Fe	26	816	242
Au	79	1422	139
Pb	82	1449	136
U	92	1535	129

with $x = k\, r_c \gtrsim 1$ across the band (e.g. Pb $x = 1.10$ at 150 keV; Fe $x = 2.07$ at 500 keV). This resolved-halo scattering constraint is the kill. *Forward-refractive channel — the moot copper bound.* The copper decrement above sits at the opposite end of x: an 8 keV probe in copper has $x = k\, r_c(\mathrm{Cu}) = 0.03$, deep in the unresolved Rayleigh regime ($x^4 \approx 1.5 \times 10^{-6}$), so the extension would add only the forward index shift $\delta_{\rm AVE} \approx 4.2 \times 10^{-8}$ quoted above — a separate, consistent, now-moot low-energy bound, not a counterexample. The exclusion is therefore a resolved-halo *scattering* constraint ($Z \gtrsim 26$, $E \gtrsim \hbar c/r_c(Z)$); the copper decrement is the unresolved low-energy *refractive* channel — opposite ends of x, no contradiction. Both channels concern a static extension the model does not assert: they are *why* it is excluded, not predictions of it.

The radiative scoping is at present an *open postulate*: it states *that* the law is radiative-sector, not yet *why*. The dynamical scoping routes proposed to derive it — keys that would separate a propagating from a held field by its dynamical content, and the weaker escape that a held static field might load the permittivity only slightly, cut off at $\ell_{\rm node}$ — have each been tested and excluded; static fields do not sit inside the model as a small effect, they sit outside it. The magnetic sector's circulation keying [Eq. (7), static-**B** transparency] is a distinct sector again and is unaffected by any of this.

C. The magnetic sector: static-flux transparency

The constitutive postulate is completed by a statement about the magnetic response, which is *not* symmetric with the electric one. The magnetic saturation factor keys on the field's *circulation* — the displacement current threading a node, $\oint \mathbf{H}\cdot d\boldsymbol{\ell} = \varepsilon_0\,\partial_t \int \mathbf{E}\cdot d\mathbf{A}$ — rather than on static flux, through the same elliptic kernel applied to a circulation amplitude A_I:

$$S_B = \sqrt{1 - A_I^2}, \quad \mu_{\text{eff}} = \frac{\mu_0}{\sqrt{1 - A_I^2}}, \quad A_I = \frac{\left|\oint \mathbf{H}\cdot d\boldsymbol{\ell}\right|}{I_{\max}}, \tag{6}$$

with $I_{\max}$ the single magnetic-sector current scale (the dual of the electric scale E_{c}, fixed by the same calibration and carrying no new parameter): the circulation one node can carry, $I_{\max} = \varepsilon_0 c\, E_{\text{c}}\,\ell_{\text{node}} = E_{\text{c}}\,\ell_{\text{node}}/Z_0 \simeq 116\,\text{A}$ (with $Z_0 = 376.73\,\Omega$ the vacuum wave impedance and $\varepsilon_0 c = 1/Z_0$; a 2π closed-loop convention gives $\simeq 730\,\text{A}$ instead — an $O(1)$ loop-geometry ambiguity immaterial to everything below). The model takes the magnetic response on the branch with the root in the *denominator*, $\mu_{\text{eff}} = \mu_0/S_B \geq \mu_0$ [Eq. (6)] — the permeability *stiffens* with circulation, the electromagnetic dual of the *softening* electric permittivity $\varepsilon_{\text{eff}} = \varepsilon_0\sqrt{1 - A^2}$ — and not the opposite-sign $\mu_{\text{eff}} = \mu_0 S_B$. At the static endpoint ($A_I = 0$) the two branches coincide at μ_0, so the static-$\mathbf{B}$ transparency of Eq. (7) is branch-independent; only the near-zone loading distinguishes them, and there the dual denominator branch is the one that pairs with the softening electric kernel. This functional has two limits that fix the magnetic phenomenology outright. *(a) Static field — the exact endpoint.* A purely static magnetic field ($\partial\mathbf{B}/\partial t = 0$) carries no circulation, so $\oint \mathbf{H}\cdot d\boldsymbol{\ell} = 0 \Rightarrow A_I = 0 \Rightarrow S_B = 1 \Rightarrow \mu_{\text{eff}} = \mu_0$, and the magnetic-route birefringence is

$$\delta n_\mu = 0 \qquad (\text{static } \mathbf{B},\ \text{exactly}). \tag{7}$$

(b) Near zone — computed PVLAS/BMV consistency. For an oscillating magnetic source the vacuum circulation is a double time-derivative of the quasistatic potential, so $A_I \propto (kr)^2$ in the near zone; a laboratory magnet modulated at Hz (PVLAS) or pulsed on the ms scale (BMV) sits at $kr \to 0$ on the optical-probe timescale, giving $A_I \to 0$, $S_B \to 1$, and $\delta n_\mu \to 0$. The existing PVLAS/BMV nulls are therefore the *computed* magnetic-sector prediction of Eq. (6), not an asserted side-condition. This is a hard, parameter-free side-prediction: the model forbids static-magnetic-field vacuum birefringence at any field strength. It is the feature that most sharply separates the model from QED, which predicts a nonzero static-$\mathbf{B}$ birefringence; exact Born–Infeld likewise predicts zero static-$\mathbf{B}$ *birefringence*, since it is the unique nonlinear electrodynamics *with a Maxwell weak-field limit* (besides Maxwell) whose two photon polarizations propagate on a single *common* effective light cone in any constant background [6, 7] — a common index shift for both polarizations ($\Delta n = 0$), not the full non-loading ($S_\mu = 1$, no index change of any kind) that Eq. (7) states for this model. (The zero-birefringence family is larger — Russo and Townsend classify it in full [4] — but the other members lack a weak-field Maxwell limit, so exact B-I is the sole physically relevant zero-Δn comparison here.) A Δn polarimeter (PVLAS/BMV) is blind to the common-shift-vs-non-loading distinction, so the zero-birefringence agreement alone does not separate the model from *exact* B-I — the electric-route coefficient (Sec. III) does. A propagating field — such as the ReLaX optical pump, for which $\mathbf{E}$ and $\mathbf{B}$ co-move with $B = E/c$ and $\partial\mathbf{B}/\partial t \neq 0$ — *does* load the response. Its magnetic contribution is nonetheless a negligible correction: the circulation the wave threads through one node is set by its own $\partial_t\mathbf{E}$, giving a circulation amplitude $A_I = A\,(k\ell_{\text{node}})$ with k the pump wavenumber, so at the $1.55\,\text{eV}$ ($\lambda \simeq 800\,\text{nm}$) carrier $k\ell_{\text{node}} \simeq 3.0\times10^{-6}$ and the magnetic loading is suppressed relative to the electric by $A_I^2/A^2 = (k\ell_{\text{node}})^2 \simeq 9\times10^{-12}$ — about eleven orders of magnitude below the electric birefringence. The electric-route result of Eq. (3) therefore already captures the full birefringence of such a wave to $\sim$1 part in 10^{11}, and the registered prediction of Table I stands on the electric sector alone. The consequences of Eq. (7) for existing null experiments are taken up in Sec. IV. Physically, the electric/magnetic asymmetry traces to the *Maxwell source asymmetry*: $\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$ has genuine charge sources, so a held electric field is a real static bias the permittivity loads on, whereas $\nabla\cdot\mathbf{B} = 0$ has no magnetic monopole, so a static magnetic field supplies no bias and the permeability loads only on circulation (∂_t, curl). The two sector keys — electric-field magnitude and magnetic circulation — are thus grounded in the single structural fact that charges exist and magnetic monopoles do not; they are *not* a consequence of the field's radiative or far-field character (a propagating wave loads because it carries both an $\mathbf{E}$ magnitude and a circulation, not because it is radiation).

D. Frame, dispersion, and the common-mode channel

Three points of self-consistency. *Frame.* Because S depends on the single quantity $|E|^2$, which is not a Lorentz invariant, the physical question is which frame the saturating medium *responds* in. The pump field magnitude is *specified* in the lab frame of the optical focus, where the laser defines it; but a real saturable dielectric keys on the field magnitude in its own rest frame, and the vacuum of this model is a material medium at rest in the substrate frame, which we identify with the cosmic (CMB) rest frame. A weak *static* laboratory field lies outside the model's scope (Sec. II, sector scope), so the apparatus's $370\,\text{km}\,\text{s}^{-1}$ motion through that frame modulates *nothing* from it — the motional field a moving magnet induces is itself quasi-static and equally out of scope, an *identical* zero rather than a small correction. The pump, however, is not static: it is a propagating radiation field, whose amplitude transforms between the lab and substrate frames by the plane-wave Doppler factor $D(\theta) = \gamma(1 + \beta\cos\theta)$ — carrying a term *first* order in $\beta = v/c$. The apparatus's motion through the substrate frame therefore does modulate the pump-driven signal, at first order, as a positive and testable prediction rather than a negligible correction — developed next.

A sidereal signature. Because the medium responds in the substrate (CMB) rest frame while the pump amplitude is fixed in the lab, the projection of the $370\,\mathrm{km\,s^{-1}}$ boost onto the pump–probe geometry varies as the Earth rotates and orbits, modulating the signal at *first* order in $\beta = v/c \simeq 1.23 \times 10^{-3}$. The pump is a radiation field, so its amplitude Doppler-shifts as $D(\theta) = \gamma(1+\beta\cos\theta)$, and to leading order the flip probability scales as $P_{\mathrm{flip}} \propto D^4$; expanding D^4, the linear-in-β term sets a fractional amplitude $4\beta \simeq 4.9 \times 10^{-3}$ at the sidereal period (one sidereal day, 86 164 s), phased to the known CMB dipole direction, with an $\sim 8\%$ annual sideband from the Earth's orbital velocity. This 4β is the pump-amplitude contribution *alone*: the same boost transforms the *probe*-side kinematics at the same first order in β — the probe-frequency Doppler shift, which enters the retardance through $\Delta\varphi \propto \omega_{\mathrm{probe}}$; the aberration of the collision angle $\vartheta_{\mathrm{coll}}$, which enters through the $(1-\cos\vartheta_{\mathrm{coll}})^4$ geometry factor; and the $O(\beta)$ change in the interaction path length — each multiplying the coefficient by a geometry-dependent $O(1)$ factor. The pump amplitude is the single largest term but does not dominate the others by an order, so we register 4β as the *order-of-magnitude* scale of the modulation rather than an exact coefficient; computing the full $O(\beta)$ coefficient requires the probe-side transforms and is deferred to the second-generation measurement in which this signature is accessed. The effect is a pure *amplitude* modulation — it rides the signal magnitude, not the polarization character or the analyzer geometry — and its *first*-harmonic (sidereal-period) character is itself a discriminator, robust to that coefficient uncertainty: the $O(\beta^2)$ second harmonic ($5\beta^2 \simeq 7.6 \times 10^{-6}$) is smaller by ~ 3 orders, so a static-field response, which would carry *only* the $O(\beta^2)$ second harmonic, is separated by the mere *presence* of the first harmonic whatever its precise amplitude. (An earlier registration quoted only that second harmonic, as $(v/c)^2 \simeq 1.5 \times 10^{-6}$, and missed the dominant first harmonic.) This is a pre-registered *third* falsifiable signature of the model, alongside the tree-level coefficient (Sec. III) and the static-$\mathbf{B}$ transparency (Sec. IV), accessible as a second-generation measurement once the leading signal is established.

This sidereal modulation is a genuine Lorentz-violating signature: a covariant theory (QED) predicts exactly zero. It is not excluded by existing Lorentz-violation bounds for two independent structural reasons. First, it is a *nonlinear* (field-dependent) response, whereas the standard photon-sector bounds — the minimal Standard-Model-Extension coefficients k_F and k_{AF} [15] — constrain only *linear*, field-independent modifications of free propagation, so there is no coefficient in that framework for a pump-induced effect. Second, no astrophysical environment pumps it: strong ambient magnetic fields are static and hence transparent (the magnetic sector keys on circulation, not flux, Eq. (7)), and the astrophysical radiation channels that could in principle pump it are either below the laboratory pump amplitude or geometrically suppressed. The geometric suppression is decisive and worth spelling out: the birefringence lives in the pump–probe cross terms and scales as $(1-\cos\vartheta_{\mathrm{coll}})^4$ with the collision angle (Sec. III), so it is maximal only for a *head-on* (counter-propagating) pump and probe — the geometry this proposal engineers deliberately — and vanishes for co-propagation. Astrophysical radiation reaching a probe photon overwhelmingly co-propagates with it ($\vartheta_{\mathrm{coll}} \to 0$), where that factor drives the response toward zero by many orders, and even a rare counter-propagating astrophysical source would still have to match the focused laboratory pump amplitude to compete. The effect is therefore unconstrained by existing observation while remaining directly testable in the counter-propagating pump–probe configuration of this proposal.

Dispersion. We treat the coefficient as frequency-independent (a DC-Kerr response), so the same E_{c} governs both the 1.55 eV pump and the $\sim$ 10 keV probe. This is justified by scale separation: the medium's natural response scale is the electron rest energy $m_e c^2 \simeq 511\,\mathrm{keV}$, so the probe sits nearly two orders of magnitude below it, $\omega_{\mathrm{probe}}/\omega_0 \simeq 0.02$. A dispersive correction enters at second order in this ratio, bounding the fractional shift of the coefficient at $(\omega_{\mathrm{probe}}/\omega_0)^2 \sim 4 \times 10^{-4}$ ($< 0.1\%$) across the probe band — negligible against the $\sim 10^7$ discrimination margin, so the quasi-static coefficient is safe. *Common-mode channel.* Beyond the birefringence, the model predicts an isotropic index shift $\delta n_{\mathrm{iso}} = \sqrt{S} - 1 \simeq -\frac{1}{4}A^2$, common to both probe eigenmodes and hence rejected by a polarimeter. It is in principle accessible to pump–probe interferometry: the DeLLight experiment [16] is a Sagnac-interferometer measurement of exactly this common-mode vacuum index change, projecting sensitivity to the QED-level induced index (a refraction angle ~ 0.13 prad, reached at 5σ in ~ 1 month with a signal-amplification extinction factor $\mathcal{F} = 0.4 \times 10^{-5}$). The present model's isotropic shift at $10^{21}\,\mathrm{W\,cm^{-2}}$ is $\sim 1.5 \times 10^{-7}$ over the focus. We do not register a prediction for the common-mode channel here; the falsifier of this Letter is the polarimetric differential, not the interferometric common mode.

III. THE PREDICTION

A. The matched differential and the readout chain

A birefringence polarimeter measures the *difference* $n_\parallel - n_\perp$ between the two eigenmodes; the common-mode index shift shared by both modes is rejected by the instrument. The falsifier observable is therefore the differential δn_{bir} of Eq. (3), and any comparison with QED must be made differential-against-differential. For a probe crossing a strong propagating field the one-loop

Euler–Heisenberg birefringence is [1, 11, 12]

$$\delta n_{\rm QED} = \frac{\alpha}{15\pi}\left(\frac{E}{E_{\rm crit}}\right)^2, \tag{8}$$

the coefficient $\alpha/(15\pi)$ being the head-on ($\vartheta_{\rm coll} = \pi$), optimal-polarization value for the counter-propagating pump–probe geometry of the BIREF@HIBEF conventional scenario [3, 11], in which the polarization-flip signal is maximized (it scales as $(1-\cos\vartheta_{\rm coll})^4$). We anchor it to two external arbiters. First, the magnetic-route differential $\delta n = 3A_eB^2$ with the measured vacuum magnetic-birefringence constant $A_e = 1.32\times10^{-24}\,\mathrm{T}^{-2}$ [8, 9]: at the $E \leftrightarrow cB$ duality point the leading $(E^2-B^2)^2$ invariant fixes the static-field (DC) coefficient at $3A_eB_{\rm crit}^2 = \alpha/(30\pi)$. The step from this static endpoint to the headline Eq. (8) is a net factor of two that factorizes into a factor of four from the head-on collision geometry (the birefringence of a probe crossing a strong pump lives in the pump–probe cross terms, not in the single on-shell plane wave, for which both Euler–Heisenberg invariants vanish), $\alpha/(30\pi) \to 2\alpha/(15\pi)$, times a $\langle\cos^2\rangle = \frac{1}{2}$ carrier average, $2\alpha/(15\pi) \to \alpha/(15\pi)$ — the static endpoint is DC, the headline is cycle-averaged, so it is not a pure geometry factor. Second, the same coefficient reproduces the focus-integrated signal estimate of the BIREF@HIBEF proposal [3] at its stated parameters. Both Eq. (3) and Eq. (8) are E^2-leading: an E^2 slope alone does *not* discriminate the two. The discriminator is the coefficient.

For a single-pass geometry (no cavity build-up), an index difference $\delta n_{\rm bir}$ over an interaction length z imprints a retardance $\Delta\varphi = (2\pi/\lambda)|\delta n_{\rm bir}|\,z$ on a probe of wavelength λ, rotating a fraction

$$P_{\rm flip} = \sin^2\left(\frac{\Delta\varphi}{2}\right) \tag{9}$$

of the probe intensity into the crossed polarimeter channel. This bounded form reduces to the perturbative $(\Delta\varphi/2)^2$ when $\Delta\varphi \ll 1$. We compute both the model and the QED co-prediction through this *identical* chain, changing only $\delta n_{\rm bir}$, so the comparison is like-for-like and free of a strawman baseline. *Field convention.* The field E entering Eqs. (3) and (8) throughout, and in Table I, is the peak carrier amplitude $E = \sqrt{2I/(c\varepsilon_0)}$ of the optical pump at intensity I; both legs are evaluated at this same peak field through the identical chain. Two distinct statements about this choice must be kept apart. Merely *re-parametrizing* the field — e.g. substituting the cycle-averaged $\langle E^2\rangle = E^2/2$ into the same two formulas — rescales both legs' absolute retardance by a common factor and leaves the ratio of Sec. III untouched. That is not the same as placing both legs' *coefficients* on a single temporal footing (instantaneous vs. cycle-averaged), which does move the numerical ratio by an $O(1)$ factor; this is the convention item ledgered in Sec. II item (iv), and it changes no order of magnitude and no falsifier verdict.

B. The field-independent ratio

Because both responses are E^2-leading, their ratio is independent of field strength. The ratio is stated on a *single temporal footing*: both coefficients are the instantaneous (peak-field) response, so the model kernel $\delta n_{\rm bir} = -\frac{1}{2}(E/E_{\rm c})^2$ is paired with the *instantaneous* one-loop coefficient $2\alpha/(15\pi)$ [the cycle-averaged headline of Eq. (8) is half of it; Sec. II item (iv)]. Using Eqs. (3), the instantaneous form of (8), and the identity (5),

$$\frac{\delta n_{\rm sat}}{\delta n_{\rm QED}} = \frac{\frac{1}{2}}{2\alpha/15\pi}\left(\frac{E_{\rm crit}}{E_{\rm c}}\right)^2 = \frac{15\pi}{4\alpha^2} = \frac{3.75\pi}{\alpha^2} \simeq 2.2\times10^5. \tag{10}$$

The model predicts a differential coefficient roughly five orders of magnitude larger than QED's, at every field. As stressed in the honesty ledger, the *value* $3.75\pi/\alpha^2$ is α-rooted — as is QED's own coefficient — and is not itself the claim; the claim is the existence of an $O(1)$ tree-level differential where QED has only a loop. (A prior draft quoted $7.5\pi/\alpha^2 \simeq 4.4\times10^5$, which paired the instantaneous model kernel against the *cycle-averaged* one-loop coefficient — a mixed footing; the single-footing value above is a factor of two smaller, and no order of magnitude or falsifier verdict changes. See Sec. II item (iv).)

C. Per-scenario predictions

Table I freezes the prediction for the scenarios of the BIREF@HIBEF proposal [3]: the probe photon energies of the conventional, dark-field, and high-energy configurations, at the demonstrated ReLaX pump intensity and at the higher design intensities, through the single-pass chain of Eqs. (9) with $z = 10\,\mu\mathrm{m}$. At the *demonstrated* pump ($10^{21}\,\mathrm{W\,cm^{-2}}$, $E \simeq 8.7\times10^{13}\,\mathrm{V\,m^{-1}}$, $A^2 \simeq 5.9\times10^{-7}$) the retardance is small ($\Delta\varphi/2 \lesssim 0.1$) and the flip probability is a well-behaved $P_{\rm flip} \simeq 4\times10^{-3}$ – -9×10^{-3} across the three probe energies — no non-perturbative regime is invoked. The QED co-prediction through the same single-pass chain (instantaneous footing, Sec. III) is $\sim 1\times10^{-13}$, so the two hypotheses are separated by ~ 11 orders of magnitude in flip probability. We note that this single-pass, flat-z chain gives a deliberately conservative QED baseline: the proposal's [3] focus-integrated estimate is $\sim 10^{-12}$ at the same pump, several orders of magnitude higher. Focus integration raises the model and QED legs by the *same* geometric factor, so the field- and geometry-independent ratio of Eq. (10) — the actual discriminator — is unchanged, and the model margin above the polarimeter floor only improves.

IV. THE SIGNATURE AND PRIOR NULLS

The model's identifying signature is that it is *electric-route active but static-magnetic-route transparent* [Eq. (7)]. Static-**B** transparency separates it cleanly

TABLE I. Frozen per-scenario predictions for the BIREF@HIBEF probe energies, computed through the single-pass readout chain [Eqs. (3), (8), (9)] at $z = 10\,\mu\text{m}$. Both legs are evaluated on the single instantaneous (peak-field) footing of Sec. III, so the QED column uses the instantaneous one-loop coefficient $2\alpha/(15\pi)$ and the model/QED column equals $(3.75\pi/\alpha^2)^2$ in the small-angle rows; the model flip probability is unchanged from the peak-field kernel. The three demonstrated-pump rows are the bankable prediction. The two design rows are shown for context: the $10^{22}\,\text{W cm}^{-2}$ row is saturating (retardance $\Delta\varphi = 1.47\,\text{rad}$, so the small-angle approximation no longer holds and the exact $\sin^2$ value is quoted), and the $10^{23}\,\text{W cm}^{-2}$ row (retardance $\Delta\varphi = 14.7\,\text{rad}$, many-wrap) is a named validity limit of the single-pass mapping and is *not* part of the registered prediction (Sec. V). The tabulated P_{flip} is the peak-field (envelope-maximum) value; a pump–probe shot integrates the flip probability over the pump's Gaussian temporal and transverse focal envelope, and because $P_{\text{flip}} \propto I^2$ in the small-angle regime the pulse-integrated expectation is the peak scaled by the fluence-weighted envelope form factor $\langle I^2\rangle/I^2_{\text{peak}} = 3^{-3/2} \simeq 0.19$ (matched co-focused Gaussian), giving $P^{\text{int}}_{\text{flip}} \simeq 1.0 \times 10^{-3}$, 8.2×10^{-4}, and 1.8×10^{-3} for the three demonstrated-pump rows — still $\sim 10^6$–10^7 above the demonstrated purity floor. (An exact fluence-weighted $\langle\sin^2\rangle$ reproduces these to $< 0.2\%$, the small-angle regime being deep.) Beyond the slow envelope, a probe crossing the $\sim 10\,\mu\text{m}$ interaction region traverses ~ 10 optical carrier cycles and so accumulates retardance from the carrier-*averaged* response rather than the crest field ($\langle\cos^2\rangle = \frac{1}{2}$ in $\delta n_{\text{bir}} \propto E^2$); folding this in scales P_{flip} by a further factor $\frac{1}{4}$ ($P_{\text{flip}} \propto \delta n^2_{\text{bir}}$), giving a *measured* expectation — envelope and carrier averaging together — of $P^{\text{meas}}_{\text{flip}} \simeq 2.6\times10^{-4}$, 2.1×10^{-4}, and 4.5×10^{-4} for the three demonstrated-pump rows, still $\sim 10^6$ above the demonstrated purity floor. The tabulated column is retained as the instantaneous peak-field (envelope-maximum) kernel of Sec. II item (iv); the model and QED legs carry identical envelope and carrier factors, so the discriminating ratio of Eq. (10) is unchanged.

Scenario (probe)	Pump [W cm^{-2}]	E [V m^{-1}]	A^2	P_{flip} (model)	P_{flip} (QED)	model/QED	regime
Conventional (9835 eV)	10^{21} (demonstrated)	8.68×10^{13}	5.90×10^{-7}	5.39×10^{-3}	1.10×10^{-13}	4.89×10^{10}	small-angle
Dark-field (8766 eV)	10^{21} (demonstrated)	8.68×10^{13}	5.90×10^{-7}	4.28×10^{-3}	8.76×10^{-14}	4.89×10^{10}	small-angle
High-energy (12 914 eV)	10^{21} (demonstrated)	8.68×10^{13}	5.90×10^{-7}	9.28×10^{-3}	1.90×10^{-13}	4.88×10^{10}	small-angle
Conventional (9835 eV)	10^{22} (design)	2.74×10^{14}	5.90×10^{-6}	4.49×10^{-1}	1.10×10^{-11}	4.07×10^{10}	saturating
Conventional (9835 eV)	10^{23} (design)	8.68×10^{14}	5.90×10^{-5}	$7.65 \times 10^{-1\dagger}$	1.10×10^{-9}	—	*beyond validity*†

† Excluded from the registered prediction: at $10^{23}\,\text{W cm}^{-2}$ the single-pass retardance is $\Delta\varphi \simeq 14.7\,\text{rad}$ (many-wrap); the naive perturbative flip probability there exceeds unity ($\simeq 54$), so the bounded $\sin^2$ value masks a broken single-pass mapping. See Sec. V.

from QED, which predicts a nonzero static-**B** vacuum birefringence that the entire PVLAS/BMV program was built to detect. It does *not*, by itself, separate the model from *exact* Born–Infeld: exact B-I is the celebrated zero-birefringence theory — among the nonlinear electrodynamics with a single effective light cone in any constant background [7], the unique one *with a Maxwell weak-field limit* [6] — so it too predicts zero static-**B** birefringence, and the two remain consistent with the same PVLAS/BMV nulls. (The zero-birefringence family is not a singleton: Russo and Townsend classify all of its members [4] — Born–Infeld, Plebański, and the reverse and extreme Born–Infeld cases, the latter three all lacking a Maxwell weak-field limit — while generic Born–Infeld-*type* family members outside that zero-birefringence set do birefringe in a static **B**. The weak-field-Maxwell qualifier is what singles out the 1934 theory here; every downstream separation below survives it, since none of the other zero-birefringence theories reduces to Maxwell at weak field.) What separates the present model from exact B-I is not the magnetic null but the electric route: the model carries a tree-level, $O(1)$ birefringence coefficient there [Sec. III], whereas exact B-I stays birefringence-free on the electric route as well. The single-cone property is proved for constant backgrounds [4, 7]; it applies to the ReLaX pump under the same locally-constant-field (geometric-optics) approximation that licenses the constant-field QED coefficient of Eq. (8) — the X-ray probe wavelength ($\sim 0.1\,\text{nm}$) is far shorter than the optical pump's spatial variation scale ($\sim 800\,\text{nm}$ carrier), so the pump is locally constant over a probe wavelength and both routes fall inside the constant-background result. On that footing exact B-I predicts a common index shift with *zero* differential birefringence for the pump, and the pump-on electric-route measurement therefore separates all three at once — the model's large tree-level signal, QED's small one-loop signal, and exact B-I's exact zero.

A consequence is that the existing magnetic-route nulls are consistent with this model *by construction*, not by smallness of a coefficient. The PVLAS experiment, in a 25-year effort, measured no vacuum magnetic birefringence at a sensitivity approaching the QED prediction [8]. In this model that null is required: $\delta n_\mu = 0$ exactly for static **B**. The magnetic-route facilities therefore neither support nor test the present prediction; they confirm the side-prediction of Eq. (7) and leave the electric route open.

The electric route requires a strong *propagating* field crossed by a polarized probe — precisely the XFEL-plus-optical-laser geometry. Theory for the QED signal in this configuration is well developed [3, 11]; the same geometry is the uniquely decisive test of the present model, because it is the one configuration in which the model and QED both predict a nonzero, matched, differential signal and differ only in coefficient.

V. FALSIFIABILITY

We commit, before any pump-on data, to the following outcome as decisive.

Kill criterion. A pump-on polarimetric run at $\gtrsim 10^{18}\,\mathrm{W\,cm^{-2}}$, with a polarimeter operating at or below the demonstrated purity floor, that measures a polarization-flip probability consistent with zero (or with the QED co-prediction) at 5σ — i.e. an upper limit $P_{\mathrm{flip}} < 10^{-8}$, still $\sim 10^6$ below the model's prediction and $\sim 10^5$ above the QED level — falsifies the electric sector of this model. This is a QED-sized differential coefficient where the model demands an $O(1)$ one, and we record it as a clean falsification: the model's electric-route hypothesis is closed, with no rescue. Conversely, a measured flip probability of the predicted size ($4 \times 10^{-3} - -9 \times 10^{-3}$ at the demonstrated pump, some seven orders of magnitude above the floor) is inconsistent with QED at this observable.

Overlap certificate (precondition for a null). A pump-on null scores as a kill only if the pump and probe are independently demonstrated to have overlapped in space and time at the stated intensity. A null recorded without such a demonstration is an *absence* of signal, not a falsification — the mirror image of a false positive: a *false execution*, in which the pump simply missed the probe and no interaction was ever driven. We therefore require, as a precondition for scoring any null as a kill, an independent overlap witness: interleaved shots on a known target that responds to the pump alone — a gas puff or a thin wire at the interaction point — producing a pump-driven signal (plasma emission, ionization yield, or pump-induced scattering) that brackets the polarimetric data runs in time and is recorded at the same nominal intensity and focal geometry. This witness certifies that the pump reached the interaction point at the stated $\gtrsim 10^{18}\,\mathrm{W\,cm^{-2}}$ independently of the crossed-channel measurement — but, keyed on the pump alone, it does not fix the femtosecond *relative* timing between pump and probe that the birefringence signal also requires. We therefore add a two-beam timing certificate: the probe's own transmission (or small-angle deflection) through the pump-driven plasma, recorded against the pump–probe delay Δt. When the pulses coincide the probe traverses the transient free-electron plasma the pump has just created and is measurably attenuated and refracted; the effect appears only inside the overlap window and vanishes outside it, so its peak position and width certify the relative arrival time to the femtosecond scale — a genuinely two-beam observable, unlike the pump-only witness. Only with both certificates present — pump-at-target and pump–probe timing — does a sub-10^{-8} crossed-channel result count as a kill; absent either, it is logged as an unwitnessed null and does not close the electric-sector hypothesis.

Bright branch: how a positive is confirmed. Registering how a positive is believed is the same discipline as registering how a null kills. A true signal at the predicted $P_{\mathrm{flip}} \simeq 4 \times 10^{-3} - -9 \times 10^{-3}$ level (Table I) is large enough to saturate an unattenuated crossed channel, so the first well-overlapped shots are taken through a calibrated attenuator in the crossed arm, sized to keep the transmitted flip signal on-scale and clear of detector saturation; the attenuation is calibrated against the pump-off direct beam so the recovered P_{flip} stays traceable. A positive is then confirmed — not merely observed — by a fingerprint battery that no pump-independent background reproduces: (i) a pump–probe delay (Δt) scan whose signal peaks at the femtosecond pump-pulse width and vanishes outside the overlap window; (ii) a $\sin^2(2\psi)$ four-lobe dependence on the angle ψ between the pump polarization and the probe analyzer, the signature of a pump-defined optical axis; and (iii) an I^2 scaling of the flip probability with pump intensity, following $\delta n_{\mathrm{bir}} \propto E^2$ [Eq. (3)] through the crossed-channel readout [$P_{\mathrm{flip}} \propto \delta n_{\mathrm{bir}}^2$, Eq. (9)], so $P_{\mathrm{flip}} \propto I^2$. A candidate that fails any one of the three is rejected as a systematic rather than accepted as a confirmation.

Why commissioning data suffices. The record X-ray-polarimeter purity is 8×10^{-11} [12], and a single-measurement demonstrated floor of 2.4×10^{-10} was reached earlier [13]; both were measured with the pump off. Taking the conservative 2.4×10^{-10} floor, at the demonstrated ReLaX intensity the model predicts $P_{\mathrm{flip}} \simeq 4 \times 10^{-3} - -9 \times 10^{-3}$ (Table I), a factor $\sim 2 \times 10^7$ above it. The prediction is thus not a precision measurement at the edge of feasibility: it is a $\sim 10^7$ effect at already-demonstrated parameters, so even a commissioning-grade pump-on polarimetric measurement adjudicates it. Figure 1 places the prediction, the QED co-prediction, the record floor, and the demonstrated pump on the exposure plane.

Systematics and reanalysis. Converting a crossed-polarimeter count ratio to a flip probability requires accounting for the instrumental factors the proposal enumerates: diamond/analyzer reflection losses ($\sim 2\%$ per reflection), lens and optics transmission, analyzer acceptance bandwidth, and beam-overlap and timing jitter, together with pump-induced backgrounds (plasma emission, scattered pump light) that could mimic or mask a signal in the crossed channel [3, 12]. Each of these is an $O(1)$–$O(10)$ correction; none approaches the $\sim 10^7$ margin between the model prediction and the floor, so the kill test is robust to the full instrumental budget. Operationally, a third party can adjudicate this prediction from a published pump-on run by extracting the crossed-channel fraction $N_\perp/N$, correcting it for the enumerated throughput and background factors, and comparing the result and its error bar against the frozen entries of Table I: a value at the 10^{-3} level confirms the model, a 5σ upper limit below 10^{-8} falsifies it.

This is a forward prediction. No published experiment has combined a strong optical pump, a polarized X-ray probe, and a polarization-flip analysis at the required geometry. The reported 8×10^{-11} [12] and 2.4×10^{-10} [13] polarimeter purities were measured with the pump off; the March-2024 HIBEF beamtime was X-ray-only, with the optical pump not fired [3]; the all-optical and X-ray–X-ray light-by-light searches, the heavy-ion measurements, and the PVLAS magnetic-route null [8] all lack the combination of a polarized X-ray probe passing through a strong optical focus with flip analysis. The exposure region that would bound the model's E-route signal is therefore empty (Fig. 1), and the present table is a pre-registration rather than a retrodiction.

An honest exclusion. The highest design intensity of the proposal, $10^{23}\,\mathrm{W\,cm^{-2}}$, drives the single-pass retardance to $\Delta\varphi \simeq 14.7\,\mathrm{rad}$ (Table I). In this many-wrap regime the crossed-polarimeter flip fraction of Eq. (9) is a rapidly oscillating function whose single-pass interpretation is not credible. We do *not* register a prediction there; it is a named validity limit of the readout mapping, not part of the falsifier. The registered prediction lives entirely at the demonstrated-pump scenarios, where the small-angle mapping holds.

Consistency with existing bounds. The same calibrated scale E_c leaves the model consistent with precision optical measurements that do not use the pump–probe geometry. The implied vacuum Kerr coefficient $n_2 = 1/(2c\varepsilon_0 E_c^2) \simeq 1.5 \times 10^{-32}\,\mathrm{m^2\,W^{-1}}$ gives a self-focusing critical power $\sim 6.5 \times 10^3\,\mathrm{PW}$, far above any existing laser, so no vacuum self-focusing is expected (a propagating-field, and hence radiative-sector, bound); the circulating fields of gravitational-wave interferometers produce a retardance $\sim 10^{-11}$ rad, far below their sensitivity (again a propagating-field bound); and static laboratory fields — weak or strong — lie outside the model's scope altogether (Sec. II, sector scope), so no static-field measurement, and no preferred-frame boost acting on one, either constrains the model or is predicted by it. Both bounds above are propagating-field (radiative-sector) bounds, the only sector the model claims. The interferometric route that would probe the common-mode index — the DeLLight experiment [16], projecting sensitivity to the QED-level index change — has not yet reported a pump-on measurement in this regime, so it too leaves the model unconstrained at present. The model is thus not already excluded by any published precision-optics bound [10, 16].

Pre-registration. The predictions of Table I are fixed before any BIREF@HIBEF pump-on collision data exist. Their priority is anchored by an OpenTimestamps proof: the SHA-256 hash of the frozen prediction is committed to the Bitcoin blockchain, so the hash (and hence the existence of a prediction fixed at that block time) is public while the manuscript content remains private until disclosure. No entry is revised after pump-on data become available, and any revised prediction would carry its own later timestamp.

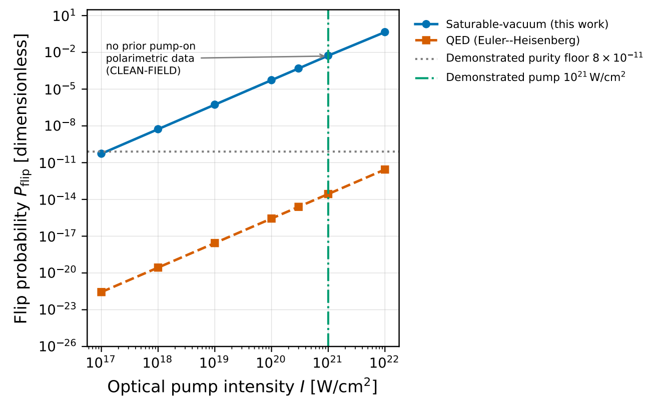

FIG. 1. The exposure plane. Predicted polarization-flip probability P_{flip} versus optical-pump intensity for the saturable-vacuum model (blue, Eqs. (3) and (9)) and for the QED Euler–Heisenberg co-prediction (orange, Eq. (8)), through the identical single-pass readout chain at probe energy 9835 eV, $z = 10\,\mu\mathrm{m}$. Both scale as intensity squared, so their ratio is field-independent. The dotted line marks the record X-ray-polarimeter purity (8×10^{-11}, Ref. [12]; the conservative 2.4×10^{-10} single-measurement floor of Ref. [13] used for the kill criterion lies just above it), and the dash-dot line the demonstrated ReLaX pump ($10^{21}\,\mathrm{W\,cm^{-2}}$). No prior experiment occupies the pump-on polarimetric exposure region, so the model's prediction there is unconstrained by existing data.

VI. CONCLUSION

Two constitutive statements — a vacuum permittivity that saturates through the elliptic kernel of Eq. (1), with one calibrated field scale, and a response that keys on the field's sources (electric-field magnitude, magnetic circulation), the sector asymmetry following from the Maxwell source structure $\nabla \cdot \mathbf{B} = 0$ — yield a definite, pre-registered, tree-level birefringence prediction for the pump–probe X-ray polarimeter geometry of BIREF@HIBEF. The prediction exceeds the QED co-prediction by the field-independent factor $3.75\pi/\alpha^2$ (both coefficients on a single instantaneous footing; Sec. III), sits $\sim 10^7$ above the demonstrated polarimeter floor at already-demonstrated pump parameters, and carries a hard side-prediction — no static-magnetic-field vacuum birefringence — that renders the existing PVLAS/BMV nulls consistent by construction and singles out the electric-route XFEL geometry as decisive. We have stated the kill criterion in advance and ledgered the one imported scale plainly. The measurement that decides the model is within reach of commissioning-grade data at an already-operating facility.

Appendix A: Derivation of the eigen-indices

The eigen-indices of Eq. (2) follow from the constitutive law Eq. (1) by direct differentiation; we display every step. The scalar law relates displacement and field through the field-magnitude invariant $u \equiv |E|^2$,

$$D_i = \varepsilon_0\, S(u)\, E_i, \qquad S(u) = \sqrt{1 - u/E_c^2}. \tag{A1}$$

A weak probe $\mathbf{e}$ rides on a strong, linearly polarized pump $\mathbf{E}_0$. Its small-signal response is the permittivity tensor $\varepsilon_{ij} = \partial D_i/\partial E_j$ evaluated at the pump. Applying the product rule to Eq. (A1),

$$\varepsilon_{ij} = \frac{\partial}{\partial E_j}\left[\varepsilon_0 S(u) E_i\right] = \varepsilon_0\Big(S\,\frac{\partial E_i}{\partial E_j} + E_i\,\frac{\partial S}{\partial E_j}\Big), \qquad \frac{\partial E_i}{\partial E_j} = \delta_{ij}, \tag{A2}$$

and the chain rule fixes the second term through $\partial S/\partial E_j = S'(u)\,\partial u/\partial E_j = 2S'(u)E_j$, so that evaluated at the pump

$$\varepsilon_{ij} = \frac{\partial}{\partial E_j}\left[\varepsilon_0 S(u) E_i\right]\Big|_{\mathbf{E}_0} = \varepsilon_0 S(u_0)\,\delta_{ij} + 2\varepsilon_0 S'(u_0)\,E_{0i}E_{0j}, \tag{A3}$$

with $u_0 = |\mathbf{E}_0|^2 \equiv E^2$ and $S' = \mathrm{d}S/\mathrm{d}u$. This is uniaxial, with optic axis along $\hat{\mathbf{E}}_0$: the two eigenvalues are $\varepsilon_0 S$ (transverse, twofold) and $\varepsilon_0(S + 2S'E^2)$ (longitudinal). Differentiating the kernel gives, with $A \equiv E/E_\mathrm{c}$,

$$S'(u) = -\frac{1}{2E_\mathrm{c}^2\sqrt{1-u/E_\mathrm{c}^2}} = -\frac{1}{2E_\mathrm{c}^2 S}, \tag{A4}$$

so that $2S'(u_0)\,E^2 = -A^2/S$. The probe indices are the square roots of the eigenvalues,

$$n_\perp = \sqrt{S} = (1-A^2)^{1/4}, \tag{A5}$$

$$n_\parallel = \sqrt{S - \frac{A^2}{S}} = \sqrt{\frac{S^2 - A^2}{S}} = \sqrt{\frac{1-2A^2}{\sqrt{1-A^2}}}, \tag{A6}$$

which are Eq. (2). Expanding for $A \ll 1$,

$$n_\perp \simeq 1 - \tfrac{1}{4}A^2, \qquad n_\parallel \simeq 1 - \tfrac{3}{4}A^2, \tag{A7}$$

so the birefringence (the polarimeter observable) and the common-mode shift (the polarimeter rejects it) are

$$\delta n_\mathrm{bir} = n_\parallel - n_\perp \simeq -\tfrac{1}{2}A^2, \qquad \delta n_\mathrm{iso} = n_\perp - 1 \simeq -\tfrac{1}{4}A^2, \tag{A8}$$

the differential being exactly twice the common-mode shift. The ratio of Sec. III then follows from Eq. (A8) and the QED coefficient alone; no parameter beyond E_c enters at any step.

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